

Cleaned Notes on Surrogate Scores and Thresholded Regression

Typeset transcription of handwritten notes

May 29, 2026

Notation convention

The handwritten notes use both a generic threshold k and a special threshold ϕ_0 . I write

$$T_k := \mathbf{1}\{Q > k\}, \quad T_0 := \mathbf{1}\{Q > \phi_0\},$$

with T_0 used when the threshold is fixed at ϕ_0 . The variables are:

Symbol	Role in the notes
X	covariates
η	latent/prognostic variable used for conditional modeling
Q	score used to define the thresholded group
$\tilde{Q}$	surrogate score or noisy/estimated version of Q
W	coefficient/response multiplying the threshold indicator
$\beta^T X$	linear covariate adjustment
$g(\eta)$	baseline adjustment depending on η
ε, r	mean-zero residual terms

The core outcome model is

$$Y = W \mathbf{1}\{Q > k\} + \beta^T X + g(\eta) + \varepsilon. \quad (1)$$

In the first handwritten page, $g(\eta)$ is absent, but it is included in the later formulation; setting $g \equiv 0$ recovers the earlier display.

1 A first conditioning calculation

The quantity of interest in the opening calculation is a centered thresholded expectation of the form

$$\mathbb{E}[(W - c)\mathbf{1}\{Q > k\}]. \quad (2)$$

Iterating expectation with respect to η gives

$$\mathbb{E}[(W - c)\mathbf{1}\{Q > k\}] = \mathbb{E}[\mathbb{E}[(W - c)\mathbf{1}\{Q > k\} \mid \eta]]. \quad (3)$$

If the conditional independence condition

$$W \perp\!\!\!\perp Q \mid \eta \quad (4)$$

holds, and if

$$\alpha(\eta) := \mathbb{E}[W \mid \eta], \quad (5)$$

then

$$\mathbb{E}[(W - c)\mathbf{1}\{Q > k\}] = \mathbb{E}[(\alpha(\eta) - c)p_k(\eta)], \quad p_k(\eta) := \mathbb{P}(Q > k \mid \eta). \quad (6)$$

Thus, under (4), the problem reduces to estimating the conditional mean $\alpha(\eta)$ and the conditional tail probability $p_k(\eta)$.

2 Using a surrogate score

One possible score representation in the notes is

$$\tilde{Q} = \delta^T X + \eta, \quad (7)$$

where $\tilde{Q}$ is a surrogate score and (δ, η) are treated as estimable from the observed surrogate-score data. The score Q takes finitely many ordered values $j \in \mathcal{J}$, a convenient working model is the multinomial or ordered-logit-type specification

$$\mathbb{P}(Q = j \mid \eta) = \frac{\exp(b_j \eta)}{\sum_{\ell \in \mathcal{J}} \exp(b_\ell \eta)}. \quad (8)$$

Then the conditional tail probability is obtained by summing over categories above the threshold:

$$\mathbb{P}(Q > k \mid \eta) = \sum_{j > k} \mathbb{P}(Q = j \mid \eta). \quad (9)$$

3 Adding the decomposition of W

The later pages introduce the decomposition

$$W = \alpha(\eta) + r, \quad (10)$$

where r is a residual term, assumed to be mean zero and independent of the relevant score randomness after conditioning on the appropriate variables. In particular, a sufficient version is

$$\mathbb{E}(r \mid X, \eta, Q) = 0. \quad (11)$$

Substituting (10) into (1) gives

$$Y = \alpha(\eta)\mathbf{1}\{Q > k\} + \beta^T X + g(\eta) + \varepsilon + r\mathbf{1}\{Q > k\}. \quad (12)$$

On the selected group $\{Q > k\}$ this reduces to

$$Y = \alpha(\eta) + \beta^T X + g(\eta) + (\varepsilon + r), \quad \text{conditional on } Q > k. \quad (13)$$

Therefore, as the notes say, “so long as the extra randomness in Q comes from independent sources other than (η, X) ,” the selected-sample regression remains a proper regression model: the residual remains conditionally mean zero.

Proposition 1 (Sufficient condition for a proper selected regression). *Suppose that*

$$\mathbb{E}(\varepsilon \mid X, \eta, Q) = 0, \quad \mathbb{E}(r \mid X, \eta, Q) = 0.$$

Then, on the event $\{Q > k\}$,

$$\mathbb{E}(\varepsilon + r \mid X, \eta, Q > k) = 0.$$

Consequently, (13) has conditionally mean-zero residual within the selected group.

Proof. The claim follows directly by iterated expectation:

$$\mathbb{E}(\varepsilon + r \mid X, \eta, Q > k) = \mathbb{E}\{\mathbb{E}(\varepsilon + r \mid X, \eta, Q) \mid X, \eta, Q > k\} = 0.$$

□

4 Why modeling Q on η alone is naive

The second page notes that a model of Q using only η can be too coarse. A richer model includes X directly:

$$\mathbb{P}(Q = j \mid X = x, \eta) = \frac{\exp(\delta_j^T x + b_j \eta)}{\sum_{\ell \in \mathcal{J}} \exp(\delta_\ell^T x + b_\ell \eta)}. \quad (14)$$

Then the threshold probability conditional only on η is obtained by averaging over the conditional distribution of X given η :

$$p_k(\eta) := \mathbb{P}(Q > k \mid \eta) = \int \mathbb{P}(Q > k \mid X = x, \eta) f_{X|\eta}(x \mid \eta) dx. \quad (15)$$

Equivalently,

$$p_k(\eta) = \int \sum_{j > k} \frac{\exp(\delta_j^T x + b_j \eta)}{\sum_{\ell \in \mathcal{J}} \exp(\delta_\ell^T x + b_\ell \eta)} f_{X|\eta}(x \mid \eta) dx. \quad (16)$$

Thus, if the parameters (δ_j, b_j) and the conditional law of X given η are estimated, $p_k(\eta)$ is also estimable. These parameters can be obtained consistently using $(\hat{\eta}, X, Q)$ values from a subsample.

5 Clarification with a deterministic linear score

The clarification pages specialize the score to a linear form. A clean version is

$$Q = \gamma^T X + \eta, \quad (17)$$

possibly with an additional independent score-noise term in a more general version. The outcome model at the special threshold ϕ_0 is

$$Y = W \mathbf{1}\{Q > \phi_0\} + \beta^T X + g(\eta) + \varepsilon. \quad (18)$$

Using $W = \alpha(\eta) + r$ gives

$$Y = \alpha(\eta) \mathbf{1}\{Q > \phi_0\} + \beta^T X + g(\eta) + \varepsilon + r \mathbf{1}\{Q > \phi_0\}. \quad (19)$$

On the treated/selected group $T_0 = 1$, this becomes

$$Y = \alpha(\eta) + \beta^T X + g(\eta) + (\varepsilon + r). \quad (20)$$

On the complementary group $T_0 = 0$, it becomes

$$Y = \beta^T X + g(\eta) + \varepsilon. \quad (21)$$

Thus the treated group identifies the combined nonparametric component $\alpha + g$, whereas the control group identifies g alone, subject to the usual regression-identifiability conditions.

6 Mean-zero condition for the residual r

The last page writes the needed condition as

$$\mathbb{E}[r \mid \mathbf{1}\{Q > \phi_0\}, X, \eta] = 0. \quad (22)$$

If $Q = \gamma^T X + \eta$ and, for simplicity, $\phi_0 = 0$, then

$$\mathbf{1}\{Q > 0\} = \mathbf{1}\{\eta > -\gamma^T X\},$$

so (22) is the same as

$$\mathbb{E}[r \mid \mathbf{1}\{\eta > -\gamma^T X\}, X, \eta] = 0. \quad (23)$$

Because the indicator is determined by (X, η) in this deterministic-score case, a sufficient condition is simply

$$\mathbb{E}(r \mid X, \eta) = 0. \quad (24)$$

A convenient way to ensure (24) is

$$r \perp\!\!\!\perp X \mid \eta \quad \text{and} \quad \mathbb{E}(r \mid \eta) = 0. \quad (25)$$

This matches the intuition in the handwritten note: conditional independence of r from X given η is a reasonable sufficient condition.

Remark 1 (Variance comparison). *Estimating $\alpha + g$ from the selected/treatment group is generally noisier than estimating g from the control group. First, the selected group may be a proper subset of the sample. Second, the residual in the selected group is $\varepsilon + r$, whereas the residual in the control group is only ε . Hence the selected-group regression can have both a smaller effective sample size and a larger residual variance.*

Action item from the handwritten note

The last handwritten page records the practical reminder:

Talk to Danny E. at 10 pm today, and for the moment put this out of my mind.